

Table 2 — DMCA Pipeline Stage Summary: ETT-h1 Case Study B  
Full 5-stage cycle: MOMENT-1-large (M\_curr) → ADWIN → TOPSIS → Quality Gate → Lag-Llama swap (Haiku copilot, SIL=1)

| Stage                | Component                                                                 | Input → Output                                                            | Key Result (ETT-h1 Case Study B)                                                                                     |
|----------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| 1<br>DT Profiling    | AAS jetson_nano<br>(inline JSON)                                          | Asset descriptor<br>→ SLA constraints                                     | SLA=100 ms, RAM=3500 MB<br>protocol=OPC-UA, sampling=1 Hz                                                            |
| 2<br>Model Selection | TOPSIS (4 criteria)<br>w_MAE=0.5, w_lat=0.3<br>w_par=0.1, w_lic=0.1       | M_adm pool (3 models)<br>→ ranked candidates                              | #1 PatchTST score=1.000<br>#2 Lag-Llama score=0.932<br>#3 Moirai score=0.000                                         |
| 3<br>Quality Gate    | 4-check validator<br>(latency / shape /<br>validity / memory)             | M_new=PatchTST<br>→ PASS/FAIL + fallback                                  | FAIL: MAE==RMSE=1.303 (Phase 1 artifact)<br>Fallback → Lag-Llama → ALL PASS                                          |
| 4<br>Drift Detection | ADWIN( $\delta$ =0.002)<br>+ DMCA-Drift v3.1.1<br>decision tree (8 feat.) | MAE stream from M_curr<br>→ detection_idx, drift_type,<br>swap_trigger    | t=12,362 (27.4% of drift window)<br>drift_type = gradual<br>swap_trigger = IMPROVEMENT (MAE ×0.46)                   |
| 5<br>Re-alignment    | Shadow deploy<br>+ Copilot L1 (Haiku)<br>+ COPILOTCONFIRM                 | M_new=Lag-Llama, SIL=1<br>→ operator alert +<br>atomic swap + audit JSONL | Shadow latency = 6.99 ms<br>Copilot alert: 323 chars (auto-approve, SIL<2)<br>Swap completed 2026-06-10T06:23:28 UTC |